

```
import tensorflow as tf
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.layers import GlobalAveragePooling2D, Dense
from tensorflow.keras.models import Model
import zipfile
import os
```

```
import kagglehub
```

```
# Download latest version
```

```
path = kagglehub.dataset_download("pranavraikokte/covid19-image-dataset")
```

```
print("Path to dataset files:", path)
```

```
Downloading to /root/.cache/kagglehub/datasets/pranavraikokte/covid19-image-dataset/2.a
100%|██████████| 158M/158M [00:01<00:00, 116MB/s]Extracting files...
```

```
Path to dataset files: /root/.cache/kagglehub/datasets/pranavraikokte/covid19-image-dat
```

```
!cp -rf /root/.cache/kagglehub/datasets/pranavraikokte/covid19-image-dataset/ver
```

```
train_dir = "Covid19-dataset/train"
```

```
test_dir = "Covid19-dataset/test"
```

```
train_datagen = ImageDataGenerator(
    rescale=1./255,
    rotation_range=20,
    zoom_range=0.2,
    horizontal_flip=True
)
```

```
test_datagen = ImageDataGenerator(
    rescale=1./255
)
```

```
train_generator = train_datagen.flow_from_directory(
    train_dir,
    target_size=(224,224),
    batch_size=32,
    class_mode='categorical'
)
```

```
test_generator = test_datagen.flow_from_directory(
    test_dir,
    target_size=(224,224),
    batch_size=32,
```

```
class_mode='categorical'  
)
```

Found 251 images belonging to 3 classes.
Found 66 images belonging to 3 classes.

```
base_model = MobileNetV2(  
    weights='imagenet',  
    include_top=False,  
    input_shape=(224,224,3)  
)  
base_model.trainable = False  
for layer in base_model.layers[-5:]:  
    layer.trainable = True  
for layer in base_model.layers[-10:]:  
    print(layer.name, layer.trainable)
```

```
block_16_expand_BN False  
block_16_expand_relu False  
block_16_depthwise False  
block_16_depthwise_BN False  
block_16_depthwise_relu False  
block_16_project True  
block_16_project_BN True  
Conv_1 True  
Conv_1_bn True  
out_relu True
```

```
x = base_model.output  
x = GlobalAveragePooling2D()(x)  
output = Dense(  
    train_generator.num_classes,  
    activation='softmax'  
) (x)  
  
model = Model(inputs=base_model.input, outputs=output)  
model.compile(  
    optimizer='adam',  
    loss='categorical_crossentropy',  
    metrics=['accuracy']  
)  
model.summary()
```


Model: "functional_2"

Layer (type)	Output Shape	Param #	Connected to
input_layer_2 (InputLayer)	(None, 224, 224, 3)	0	-
Conv1 (Conv2D)	(None, 112, 112, 32)	864	input_layer_2[0]...
bn_Conv1 (BatchNormalizatio...	(None, 112, 112, 32)	128	Conv1[0][0]
Conv1_relu (ReLU)	(None, 112, 112, 32)	0	bn_Conv1[0][0]
expanded_conv_dept... (DepthwiseConv2D)	(None, 112, 112, 32)	288	Conv1_relu[0][0]
expanded_conv_dept... (BatchNormalizatio...	(None, 112, 112, 32)	128	expanded_conv_de...
expanded_conv_dept... (ReLU)	(None, 112, 112, 32)	0	expanded_conv_de...
expanded_conv_proj... (Conv2D)	(None, 112, 112, 16)	512	expanded_conv_de...
expanded_conv_proj... (BatchNormalizatio...	(None, 112, 112, 16)	64	expanded_conv_pr...
block_1_expand (Conv2D)	(None, 112, 112, 96)	1,536	expanded_conv_pr...
block_1_expand_BN (BatchNormalizatio...	(None, 112, 112, 96)	384	block_1_expand[0]...
block_1_expand_relu (ReLU)	(None, 112, 112, 96)	0	block_1_expand_B...
block_1_pad (ZeroPadding2D)	(None, 113, 113, 96)	0	block_1_expand_r...
block_1_depthwise (DepthwiseConv2D)	(None, 56, 56, 96)	864	block_1_pad[0][0]
block_1_depthwise_... (BatchNormalizatio...	(None, 56, 56, 96)	384	block_1_depthwis...
block_1_depthwise_... (ReLU)	(None, 56, 56, 96)	0	block_1_depthwis...
block_1_project (Conv2D)	(None, 56, 56, 24)	2,304	block_1_depthwis...
block_1_project_BN (BatchNormalizatio...	(None, 56, 56, 24)	96	block_1_project[...
block_2_expand (Conv2D)	(None, 56, 56, 144)	3,456	block_1_project_...

block_2_expand_bn (BatchNormalizatio...	(None, 56, 56, 144)	576	block_2_expand[0...
block_2_expand_relu (ReLU)	(None, 56, 56, 144)	0	block_2_expand_B...
block_2_depthwise (DepthwiseConv2D)	(None, 56, 56, 144)	1,296	block_2_expand_r...
block_2_depthwise_... (BatchNormalizatio...	(None, 56, 56, 144)	576	block_2_depthwis...
block_2_depthwise_... (ReLU)	(None, 56, 56, 144)	0	block_2_depthwis...
block_2_project (Conv2D)	(None, 56, 56, 24)	3,456	block_2_depthwis...
block_2_project_BN (BatchNormalizatio...	(None, 56, 56, 24)	96	block_2_project[...
block_2_add (Add)	(None, 56, 56, 24)	0	block_1_project_... block_2_project_...
block_3_expand (Conv2D)	(None, 56, 56, 144)	3,456	block_2_add[0][0]
block_3_expand_BN (BatchNormalizatio...	(None, 56, 56, 144)	576	block_3_expand[0...
block_3_expand_relu (ReLU)	(None, 56, 56, 144)	0	block_3_expand_B...
block_3_pad (ZeroPadding2D)	(None, 57, 57, 144)	0	block_3_expand_r...
block_3_depthwise (DepthwiseConv2D)	(None, 28, 28, 144)	1,296	block_3_pad[0][0]
block_3_depthwise_... (BatchNormalizatio...	(None, 28, 28, 144)	576	block_3_depthwis...
block_3_depthwise_... (ReLU)	(None, 28, 28, 144)	0	block_3_depthwis...
block_3_project (Conv2D)	(None, 28, 28, 32)	4,608	block_3_depthwis...
block_3_project_BN (BatchNormalizatio...	(None, 28, 28, 32)	128	block_3_project[...
block_4_expand (Conv2D)	(None, 28, 28, 192)	6,144	block_3_project_...
block_4_expand_BN (BatchNormalizatio...	(None, 28, 28, 192)	768	block_4_expand[0...
block_4_expand_relu (ReLU)	(None, 28, 28, 192)	0	block_4_expand_B...
block_4_depthwise (DepthwiseConv2D)	(None, 28, 28, 192)	1,728	block_4_expand_r...

block_4_depthwise_... (BatchNormalizatio...	(None, 28, 28, 192)	768	block_4_depthwis...
block_4_depthwise_... (ReLU)	(None, 28, 28, 192)	0	block_4_depthwis...
block_4_project (Conv2D)	(None, 28, 28, 32)	6,144	block_4_depthwis...
block_4_project_BN (BatchNormalizatio...	(None, 28, 28, 32)	128	block_4_project[...
block_4_add (Add)	(None, 28, 28, 32)	0	block_3_project_... block_4_project_...
block_5_expand (Conv2D)	(None, 28, 28, 192)	6,144	block_4_add[0][0]
block_5_expand_BN (BatchNormalizatio...	(None, 28, 28, 192)	768	block_5_expand[0...
block_5_expand_relu (ReLU)	(None, 28, 28, 192)	0	block_5_expand_B...
block_5_depthwise (DepthwiseConv2D)	(None, 28, 28, 192)	1,728	block_5_expand_r...
block_5_depthwise_... (BatchNormalizatio...	(None, 28, 28, 192)	768	block_5_depthwis...
block_5_depthwise_... (ReLU)	(None, 28, 28, 192)	0	block_5_depthwis...
block_5_project (Conv2D)	(None, 28, 28, 32)	6,144	block_5_depthwis...
block_5_project_BN (BatchNormalizatio...	(None, 28, 28, 32)	128	block_5_project[...
block_5_add (Add)	(None, 28, 28, 32)	0	block_4_add[0][0... block_5_project_...
block_6_expand (Conv2D)	(None, 28, 28, 192)	6,144	block_5_add[0][0]
block_6_expand_BN (BatchNormalizatio...	(None, 28, 28, 192)	768	block_6_expand[0...
block_6_expand_relu (ReLU)	(None, 28, 28, 192)	0	block_6_expand_B...
block_6_pad (ZeroPadding2D)	(None, 29, 29, 192)	0	block_6_expand_r...
block_6_depthwise (DepthwiseConv2D)	(None, 14, 14, 192)	1,728	block_6_pad[0][0]
block_6_depthwise_... (BatchNormalizatio...	(None, 14, 14, 192)	768	block_6_depthwis...
block_6_depthwise_...	(None, 14, 14, 192)	0	block_6_depthwis...

(ReLU)	192)		
block_6_project (Conv2D)	(None, 14, 14, 64)	12,288	block_6_depthwis...
block_6_project_BN (BatchNormalizatio...	(None, 14, 14, 64)	256	block_6_project[...
block_7_expand (Conv2D)	(None, 14, 14, 384)	24,576	block_6_project_...
block_7_expand_BN (BatchNormalizatio...	(None, 14, 14, 384)	1,536	block_7_expand[0...
block_7_expand_relu (ReLU)	(None, 14, 14, 384)	0	block_7_expand_B...
block_7_depthwise (DepthwiseConv2D)	(None, 14, 14, 384)	3,456	block_7_expand_r...
block_7_depthwise_... (BatchNormalizatio...	(None, 14, 14, 384)	1,536	block_7_depthwis...
block_7_depthwise_... (ReLU)	(None, 14, 14, 384)	0	block_7_depthwis...
block_7_project (Conv2D)	(None, 14, 14, 64)	24,576	block_7_depthwis...
block_7_project_BN (BatchNormalizatio...	(None, 14, 14, 64)	256	block_7_project[...
block_7_add (Add)	(None, 14, 14, 64)	0	block_6_project_... block_7_project_...
block_8_expand (Conv2D)	(None, 14, 14, 384)	24,576	block_7_add[0][0]
block_8_expand_BN (BatchNormalizatio...	(None, 14, 14, 384)	1,536	block_8_expand[0...
block_8_expand_relu (ReLU)	(None, 14, 14, 384)	0	block_8_expand_B...
block_8_depthwise (DepthwiseConv2D)	(None, 14, 14, 384)	3,456	block_8_expand_r...
block_8_depthwise_... (BatchNormalizatio...	(None, 14, 14, 384)	1,536	block_8_depthwis...
block_8_depthwise_... (ReLU)	(None, 14, 14, 384)	0	block_8_depthwis...
block_8_project (Conv2D)	(None, 14, 14, 64)	24,576	block_8_depthwis...
block_8_project_BN (BatchNormalizatio...	(None, 14, 14, 64)	256	block_8_project[...
block_8_add (Add)	(None, 14, 14, 64)	0	block_7_add[0][0... block_8_project_...

block_9_expand (Conv2D)	(None, 14, 14, 384)	24,576	block_8_add[0][0]
block_9_expand_BN (BatchNormalizatio...	(None, 14, 14, 384)	1,536	block_9_expand[0...
block_9_expand_relu (ReLU)	(None, 14, 14, 384)	0	block_9_expand_B...
block_9_depthwise (DepthwiseConv2D)	(None, 14, 14, 384)	3,456	block_9_expand_r...
block_9_depthwise_... (BatchNormalizatio...	(None, 14, 14, 384)	1,536	block_9_depthwis...
block_9_depthwise_... (ReLU)	(None, 14, 14, 384)	0	block_9_depthwis...
block_9_project (Conv2D)	(None, 14, 14, 64)	24,576	block_9_depthwis...
block_9_project_BN (BatchNormalizatio...	(None, 14, 14, 64)	256	block_9_project[...
block_9_add (Add)	(None, 14, 14, 64)	0	block_8_add[0][0]... block_9_project_...
block_10_expand (Conv2D)	(None, 14, 14, 384)	24,576	block_9_add[0][0]
block_10_expand_BN (BatchNormalizatio...	(None, 14, 14, 384)	1,536	block_10_expand[...
block_10_expand_re... (ReLU)	(None, 14, 14, 384)	0	block_10_expand_...
block_10_depthwise (DepthwiseConv2D)	(None, 14, 14, 384)	3,456	block_10_expand_...
block_10_depthwise... (BatchNormalizatio...	(None, 14, 14, 384)	1,536	block_10_depthwi...
block_10_depthwise... (ReLU)	(None, 14, 14, 384)	0	block_10_depthwi...
block_10_project (Conv2D)	(None, 14, 14, 96)	36,864	block_10_depthwi...
block_10_project_BN (BatchNormalizatio...	(None, 14, 14, 96)	384	block_10_project...
block_11_expand (Conv2D)	(None, 14, 14, 576)	55,296	block_10_project...
block_11_expand_BN (BatchNormalizatio...	(None, 14, 14, 576)	2,304	block_11_expand[...
block_11_expand_re... (ReLU)	(None, 14, 14, 576)	0	block_11_expand_...
block_11_depthwise (DenthwiseConv2D)	(None, 14, 14, 576)	5,184	block_11_expand_...

block_11_depthwise...	(None, 14, 14, 576)	2,304	block_11_depthwi...
block_11_depthwise... (ReLU)	(None, 14, 14, 576)	0	block_11_depthwi...
block_11_project (Conv2D)	(None, 14, 14, 96)	55,296	block_11_depthwi...
block_11_project_BN (BatchNormalizatio...	(None, 14, 14, 96)	384	block_11_project...
block_11_add (Add)	(None, 14, 14, 96)	0	block_10_project... block_11_project...
block_12_expand (Conv2D)	(None, 14, 14, 576)	55,296	block_11_add[0][...
block_12_expand_BN (BatchNormalizatio...	(None, 14, 14, 576)	2,304	block_12_expand[...
block_12_expand_re... (ReLU)	(None, 14, 14, 576)	0	block_12_expand_...
block_12_depthwise (DepthwiseConv2D)	(None, 14, 14, 576)	5,184	block_12_expand_...
block_12_depthwise... (BatchNormalizatio...	(None, 14, 14, 576)	2,304	block_12_depthwi...
block_12_depthwise... (ReLU)	(None, 14, 14, 576)	0	block_12_depthwi...
block_12_project (Conv2D)	(None, 14, 14, 96)	55,296	block_12_depthwi...
block_12_project_BN (BatchNormalizatio...	(None, 14, 14, 96)	384	block_12_project...
block_12_add (Add)	(None, 14, 14, 96)	0	block_11_add[0][... block_12_project...
block_13_expand (Conv2D)	(None, 14, 14, 576)	55,296	block_12_add[0][...
block_13_expand_BN (BatchNormalizatio...	(None, 14, 14, 576)	2,304	block_13_expand[...
block_13_expand_re... (ReLU)	(None, 14, 14, 576)	0	block_13_expand_...
block_13_pad (ZeroPadding2D)	(None, 15, 15, 576)	0	block_13_expand_...
block_13_depthwise (DepthwiseConv2D)	(None, 7, 7, 576)	5,184	block_13_pad[0][...
block_13_depthwise... (BatchNormalizatio...	(None, 7, 7, 576)	2,304	block_13_depthwi...
block_13_depthwise...	(None, 7, 7, 576)	0	block_13_depthwi...

block_13_depthwise... (ReLU)	(None, 7, 7, 370)	0	block_13_depthwi...
block_13_project (Conv2D)	(None, 7, 7, 160)	92,160	block_13_depthwi...
block_13_project_BN (BatchNormalizatio...	(None, 7, 7, 160)	640	block_13_project...
block_14_expand (Conv2D)	(None, 7, 7, 960)	153,600	block_13_project...
block_14_expand_BN (BatchNormalizatio...	(None, 7, 7, 960)	3,840	block_14_expand[...
block_14_expand_re... (ReLU)	(None, 7, 7, 960)	0	block_14_expand_...
block_14_depthwise (DepthwiseConv2D)	(None, 7, 7, 960)	8,640	block_14_expand_...
block_14_depthwise... (BatchNormalizatio...	(None, 7, 7, 960)	3,840	block_14_depthwi...
block_14_depthwise... (ReLU)	(None, 7, 7, 960)	0	block_14_depthwi...

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